Linear Programming Approach to the Graph Square Problem

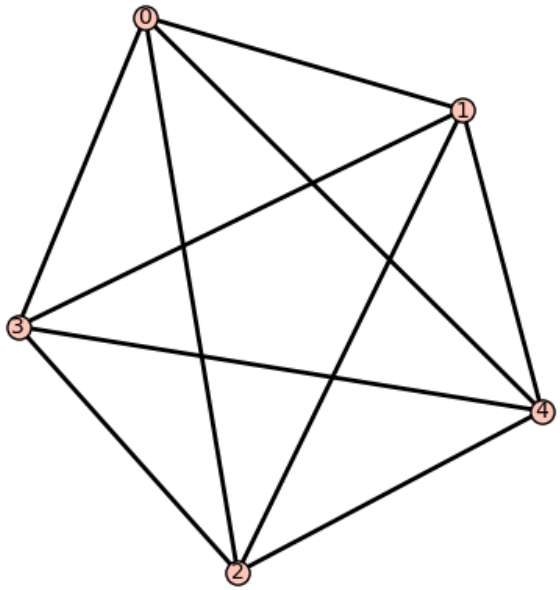
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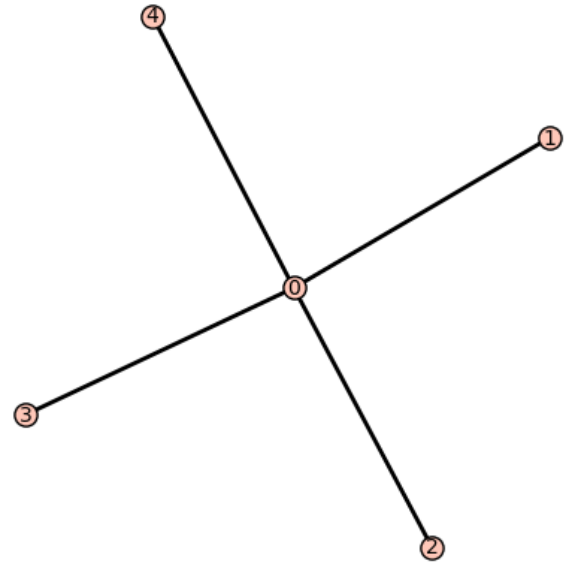
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Square of a Graph



(a) K_5.



(b) $K_{1,4}$, a root of K_5.

Figure: One of the 15 non-isomorphic graph roots of K_5.

The Graph Square Problem

Restricting on the Square vs the Root

NP-Completeness

Maximal Cliques and Unicliqual Vertices in Tree Roots

Integer Program

Linear Relaxation

Min and Max Objective Function

Identifying Square Characteristics of Tree Roots

Leaf Fixing Linear Program

Propagation of the Fixing

Vertex Deletion

Symbolic Sets

Duality